Environmental drivers of low vaccine responsiveness in a lab-to-wild mouse model

Simon A. Babayan[1,†,*], Saudamini Venkatesan[2,†], Jessica Hall[1,2], Ewan Smith[1], Amy Sweeny[2], and Amy B. Pedersen[2,*]

[1]School of Biodiversity, One Health & Veterinary Medicine, University of Glasgow, Glasgow, UK
[2]Institute of Ecology and Evolution, School of Biological Sciences, University of Edinburgh, Edinburgh, UK
[†]These authors contributed equally to this work
[*]Corresponding authors: simon.babayan@glasgow.ac.uk, amy.pedersen@ed.ac.uk

10th January 2024

TODO LIST

ABSTRACT

Vaccination is the most effective way to prevent infectious disease and protect public health. However, most new vaccines fail late into the clinical trail pipeline, and even established vaccines often preform worse in populations that differ substantially from those in which they were initially tested, such as rural *versus* urban individuals. This can lead to breakthrough infections and the inability to achieve or maintain herd immunity. However, the causes of vaccine hyporesponsiveness remain difficult to identify, quantify, and therefore pre-empt. We hypothesise that the leading causes of vaccine hyporesponsiveness are environmental, and are exacerbated when individuals are required to allocate limited resources to growth, reproduction and immunity. Using paired cohorts of laboratory-reared and wild wood mice (*Apodemus sylvaticus*) vaccinated with a single and double-dose diphtheria toxoid vaccine (with adjuvant alum), we show that circulating vaccine-specific antibodies dropped on average by 50% in a wild population relative to individuals of the same species that had been reared in laboratory conditions. We also found that, independently of habitat, substantial variation in vaccine responsiveness was explained by diet, with weaker negative impacts of being male and reproductively active. Surprisingly, helminth infection did not directly suppress vaccine-specific antibody production once adjusted for habitat, diet, and sex. While much variation remains to be explained, our results indicate that the environment plays a dramatic role in vaccine failure and that laboratory settings may systematically overestimate vaccine performance by failing to capture natural environmental variability. We anticipate that our study inaugurates a robust modelling approach and biological system for disentangling and quantifying the complex factors that cause vaccine responsiveness in diverse target populations. It may also call for a need to systematically raise efficacy thresholds for initiating clinical phases of vaccine development, and suggests that simple dietary interventions could improve individual responsiveness to immunisation.

INTRODUCTION

Unanticipated variability in individuals' responses to vaccination can lead to a lack of protection from the targetted pathogen, continued disease transmission, and an increased risk of vaccine escape[1–4]. This heterogeneity is largely driven by the individual's pre-existing immune state, itself the outcome of complex interactions between intrinsic and extrinsic factors. Indeed, vaccine efficacy can be affected by genetics[5], sex[6], reproductive status[7], past and current infections[8–10], and nutrition[11–13]. These heterogeneities can combine to affect both adaptive and innate immunity as demonstrated by the association between innate inflammatory markers in patients' peripheral blood before vaccination and subsequent antibody responses to 13 different vaccines[14], thereby potentially driving disease and transmission[15] and manifesting in the substantial drop in vaccine immunogenicity and efficacy in rural relative to urban populations[16]. However, it is unclear how these factors interact and how to mitigate their effects to maximise vaccination efficacy. The limitations to our understanding of variability in vaccine efficacy stem in part from the inadequacy of current pre-clinical models used in vaccine development, which typically utilise defined inbred, non-reproductive, single sex, age-controlled animals that are bred in captivity under strictly controlled pathogen-free conditions. As a result, we hypothesise that this might explain why only a small minority of vaccines successfully transition from pre-clinical stages of development through phase III trials and reach mass market[17] – in short, a generalisability and a transportability problem. Moreover, even successful vaccines can suffer from substantial variability between individuals and groups, notably for people of different ancestry[18] or different habitats[16]. For example, individuals of Tanzanian ancestry were reported to have higher inflammatory and type 1 immune markers[19] and responses to measles vaccination[20] than those of European ancestry, while in rural populations the expression of BCG- and tetanus-specific cytokine recall and antibody responses was found to be markedly lower than in urban populations, while humoral responses to tetanus vaccination were higher[21,22]. Taken together, these reports indicate that vaccine efficacy depends on both the vaccine and the recipients' traits, and thus a need to develop flexible and adaptable models that can be used to predict vaccine efficacy in different populations.

To increase the generalisability of experimental models of immunology and vaccinology, recent efforts have been made to integrate natural variability into immunological research[23–27]. These include exposing mice to richer, "dirtier" environments, littermates, and microbiota, which results in mice developing more mature, and thus more representative, immune systems and responses to infection. However, there is still uncertainty about how this heterogeneity affects the innate and adaptive arms of the immune system[28], and what the consequences might be for vaccine responsiveness. More specifically, it is unclear how environmental factors and past stressors influence organisms' responses to immunisation. This complexity blurs the causal relationships between environmental and immunological processes, making the design and interpretation of *in vivo* vaccination experiments difficult and their impact on their intended target populations intractable.

Here we propose a structural causal model (SCM)[29] to predict the effects of habitat, diet, sex, and parasite infection on vaccine responsiveness and to identify the paths through which their effects are mediated. This causal framework, which utilises graphical models to inform adjustment strategies[30,31], can eliminate or reduce multiple sources of confounding, such as common causes (misattribution of causal effects), sampling biases (non-representative test population), and transportability biases (lack of generalisability). We performed a vaccination study using a coupled laboratory-wild wood mouse (*Apodemus sylvaticus*) system, which allows longitudinal sampling as well as experimental manipulation in both controlled laboratory conditions and in their natural woodland habitat. This allowed us to use the same assays and interventions across habitats, thereby approximating a model of the lab-to-trial vaccine testing or to the urban-rural divide. We then compared the generation of vaccine-specific antibody responses to an alum-adjuvanted diphtheria toxoid vaccine (DTV) between laboratory and natural settings and between two diets (supplemented and control) after vaccination. We have previously demonstrated that this high quality, whole diet supplementation drives a sustained reduction in helminth (*Heligmosomoides polygyrus*) burdens and transmission, while also improving the efficacy of anthelmintic treatment[32]. We therefore predicted that vaccine responses in wild mice would be lower than in lab wood mice and higher in diet-supplemented than normal-diet animals.

To identify and quantify the drivers of vaccine responsiveness in laboratory and wild settings, we used Bayesian models to identify causal relationships defined by the SCM and thereby quantify the physiological and environmental direct and indirect causal effects of variation in the ability of recipients to generate vaccine-specific antibody responses (vaccine responsiveness). This allowed us to identify the main causes of loss of vaccine efficacy when translating a vaccine from the laboratory to the wild, and to accurately model how uncertainty and biological variability propagate through the causal network (see casual flow diagram, Fig. S1).

RESULTS

Vaccination induced stronger but variable antibody responses in the laboratory than in the wild

We conducted a paired cross-factorial study on wild and laboratory *A. sylvaticus* mice to evaluate the impact of the laboratory environment, diet supplementation, and vaccination protocol on vaccine responsiveness. The mice were given either a high or low-quality diet and then immunised with a diphtheria-alum vaccine (DTV) once or twice (see details in Materials, Methods, and Models; Figure 1).

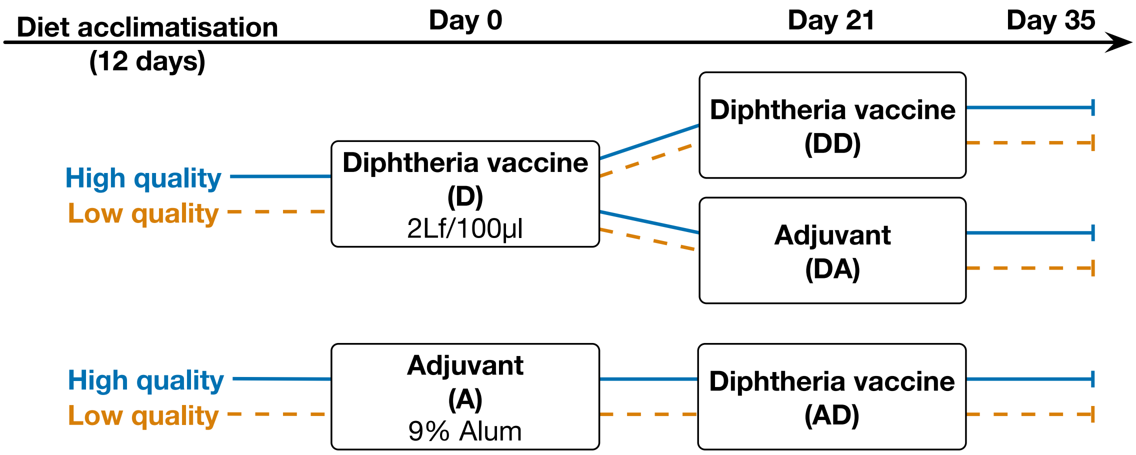


Fig. 1: Allocation of diet and vaccines. Wood mice from a laboratory colony, and from the wild were provided enriched (TransBreed) or standard (lab: normal chow; wild: no supplementation) diets. After at least 12 days, mice were allocated at random to immunisation with a 2Lf/100µl Diphtheria Toxoid + 9% alum vaccine (D) or with adjuvant alone (A). Twenty-one days after their first immunisation, mice that had been immunised with D were then given a second shot of the vaccine (DD) or adjuvant alone (DA), while mice that had received an initial adjuvant vaccine (A) were all given D for their second shot (AD).

Wild mice generated 46.9% $\pm$ 0.1 % lower DTV-specific IgG1 antibody responses than their lab-based conspecifics, irrespective of vaccine regimen or diet (Figure 2A, 2B 'Hab'). While immunisation was most effective with two doses in both habitats, lab mice reached peak antibody levels with only a single dose (Figure 2A, 'DA' vs 'DD'; Lab boosted vs. Lab non-boosted OD = 0.2 $\pm$ 0.1 whereas Wild boosted vs. Wild non-boosted OD = 1.02 $\pm$ 0.23). However, in the wild, only when given two doses did the mice show responsiveness similar to the level observed in the laboratory mice (Figure 2A, 'DD'; difference between Wild non-boosted vs. Lab non-boosted OD = 0.75 $\pm$ 0.11 while difference between Wild boosted vs. Lab boosted OD = 0.61 $\pm$ 0.18; LRT of Habitat $\times$vaccine regimen: $\Delta dof = 4, \chi^2 = 30.6, p < 0.001$). Only mice that were administered the DT antigen generated DTV-specific IgG1, indicating that there was no pre-existing immunity to this antigen in our study populations (Figure 2A, group 'A').

To further quantify how habitat (lab *vs.* wild), diet (supplemented *vs.* control), and the immunisation protocol were associated with variation in vaccine responsiveness, we constructed a multilevel Bayesian model of those factors with intercepts for each immunisation

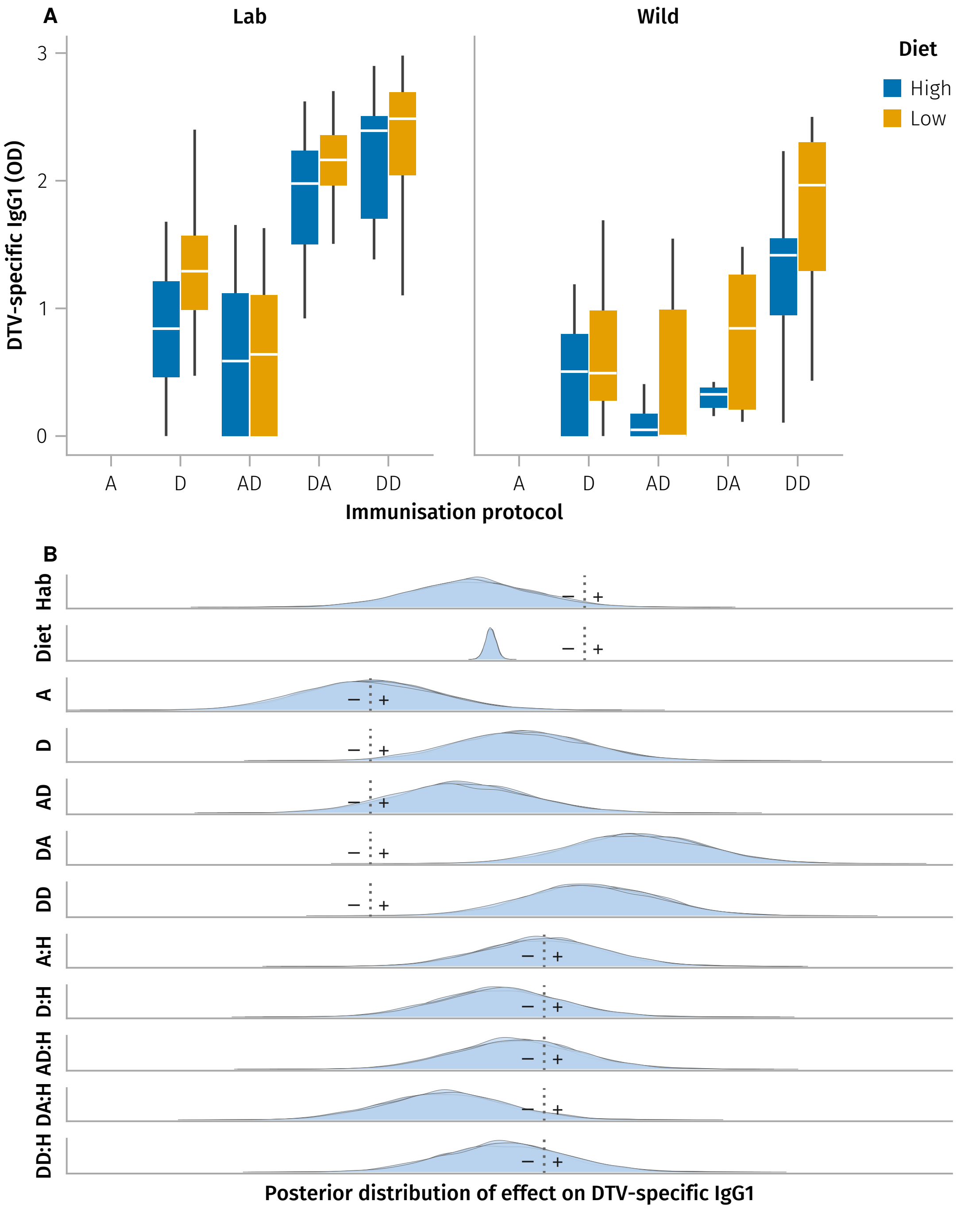


Fig. 2: Vaccine-induced DTV-specific IgG1. A, diphtheria toxoid vaccine (DTV)-specific IgG1 optical density in mice from laboratory- and wild-habitats offered high and low quality dietary supplements. Mice were immunised either once, with adjuvant as control (A) or DTV (D) only, or twice (i.e. with booster), using adjuvant alone followed by DTV at subsequent recapture ≥7 days after the previous immunisation (AD), DTV and subsequently adjuvant alone (DA), or two separate doses of DTV (DD). Only mice immunised at least 7 days prior to antibody measurement are shown, and each point represents one of 1–4 recaptures for each individual. **B**, distribution of posterior distributions of the coefficients for diet, habitat, and vaccine formulation (A, D, AD, DA, DD), and effect of the same vaccines conditional on being in the wild habitat (H) of a multilevel Bayesian model adjusted for repeated individual measures. Each density curve represents the distribution of 3,000 samples from one of four MCMC chains; dotted line represents the the value for which the reference (global intercept for Diet and Habitat; Adjuvant alone for the vaccines) is null (see parameter estimates in Table 1).

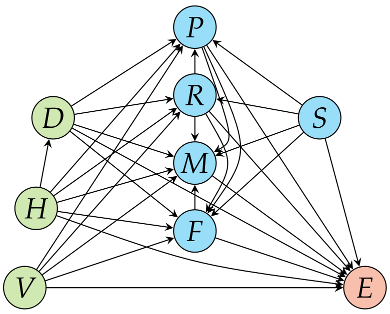
protocol (Figure 2B, Table 1). Following model selection, we fit a model with mouse Id as random effect, an interaction between habitat and vaccine formulation, and main effects of diet, habitat, and vaccine. This model indicated that the effects of each vaccine formulation on DTV-specific IgG1 OD were substantially more variable than those of habitat and diet. Further, it shows that vaccine responsiveness for all regimens but especially DA, i.e. a single immunisation, was substantially degraded by the wild, indicating a more rapid loss of vaccine-induced immunity in the wild than in the lab (Figure 2B, Table 1). However, while this model describes associations between habitat, sex, diet, and vaccine responsiveness, it does not point to which mechanisms might be targeted to prevent this loss of responsiveness. Further, it does not account for potential confounding and mediating effects of sex, reproductive status, body mass, body fat, or parasite burdens, which we hypothesise to contribute to variation in vaccine responsiveness.

Construction of a structural causal model of vaccine responsiveness

Next, to understand more specifically the relative contributions of diet and habitat to vaccine responsiveness, and how that effect might be mediated by parasite infection, reproductive status, and body condition, we built a causally-explicit model of the hypothesised causal relationships between (Figure 3).

To estimate vaccine responsiveness E as a function of vaccination V, habitat H and diet supplementation D, and more importantly, to identify which variables might act as confounders in need of adjustment or mediators that might modulate vaccine responsiveness, we constructed a directed acyclic graph (DAG, Figure 3A) to represent the hypothesised structure of the biological processes that generate the data observed in this system. We used this DAG to specify the structural causal model $\mathfrak{C}$ (Figure 3B) that links our variables of interest to vaccine efficacy E and describes the flow of causal effects based on the following criteria. Vaccination V being allocated at random, has no parent node. Its effects were hypothesised to be affected by habitat H since date of capture in the wild was inherently

A



B

$$\begin{aligned} V &:= \mathcal{U}_V \\ H &:= \mathcal{U}_H \\ S &:= \mathcal{U}_S \\ D &:= \beta_{HD}H + \mathcal{U}_D \\ R &:= \beta_{DR}D + \beta_{HR}H + \beta_{VR}V + \beta_{SR}S + \mathcal{U}_R \\ P &:= \beta_{TP}V + \beta_{DP}D + \beta_{HP}H + \beta_{RP}R + \beta_{SP}S + \mathcal{U}_P \\ F &:= \beta_{TF}V + \beta_{DF}D + \beta_{HF}H + \beta_{SF}S + \beta_{RF}R + \beta_{PF}P + \mathcal{U}_F \\ M &:= \beta_{VM}V + \beta_{HM}H + \beta_{SM}S + \beta_{RM}R + \beta_{PM}P + \beta_{FM}F + \\ &\quad \beta_{DM}D + \mathcal{U}_M \\ E &:= \beta_{VE}V + \beta_{HE}H + \beta_{FE}F + \beta_{DE}D + \beta_{ME}M + \beta_{RE}R + \\ &\quad \beta_{PE}P + \beta_{SE}S + \mathcal{U}_E \end{aligned} \tag{1}$$

Fig. 3: Graphical Causal Model of vaccine efficacy. A, Directed acyclic graph representing hypothesised causal effects expressed by the SCM $\mathfrak{C}$ (Model 1, right) driving variation in efficacy E of vaccine V in laboratory or wild mice H, under supplemented or control diet D (green vertices depict those that were experimentally intervened upon). Covariates and potential confounders (blue vertices) were time since vaccination T, parasite infection (presence or burden) P, reproductive status R, body mass M, fat scores F, and sex S. Directed edges depict causal effects hypothesised to operate between vertices. **B**, Assignments derived from the DAG with coefficients β_{ij} representing the effect of cause i on effect j. $\mathcal{U}_j$ represents the unmeasured variation of the response variable, not represented in the DAG panel A for clarity.

more variable than date of procedure in the lab for logistical reasons. A mouse being born in the lab or in the wild habitat H was considered random, and therefore took no causal input but was hypothesised to potentially affect all other variables except V and sex S. Diet D was allocated at random in the laboratory colony, but because wild mice could also forage freely, we considered that being born in the wild could have a causal effect on diet $H \rightarrow D$, specifically on the effect of the supplementation relative to the control laboratory or wild diet. Diet was hypothesised to affect reproductive status R, body mass M, body fat F and parasite burdens P as well as E, with fat scores F hypothesised to be a cause of variation in body mass M. Sex S being genetically determined and therefore subject to Mendelian randomisation[33], and all treatments being randomised with respect to sex, S was considered causally prior to reproductive status R, body mass M, body fat F and parasite burdens P. Reproductive status R being driven by sex, it was considered causally prior to body mass M and parasite burdens P. Vaccine efficacy E was hypothesised to be potentially affected by all other variables. Each node also had its own error term to capture all unobserved sources of variation, assumed to be i.i.d To estimate vaccine responsiveness E as a function of vaccination V, habitat H and diet supplementation D, and more importantly, to identify which variables might act as confounders in need of adjustment or mediators that might modulate vaccine responsiveness, we constructed a directed acyclic graph (DAG, Figure 3A) to represent the hypothesised structure of the biological processes that generate the data observed in this system. We used this DAG to specify the structural causal model $\mathfrak{C}$ (Figure 3B) that links our variables of interest to vaccine efficacy E and describes the flow of causal effects based on the following criteria. Vaccination V being allocated at random, has no parent node. Its effects were hypothesised to be affected by habitat H since date of capture in the wild was inherently more variable than date of procedure in the lab for logistical reasons. A mouse being born in the lab or in the wild habitat H was considered random, and therefore took no causal input but was hypothesised to potentially affect all other variables except V and sex S. Diet D was allocated at random in the laboratory colony, but because wild mice could also forage freely, we considered that being born in the wild could have a causal effect on diet $H \rightarrow D$, specifically on the effect of the supplementation relative to the control laboratory or wild diet. Diet was hypothesised to affect reproductive status R, body mass M, body fat F and parasite burdens P as well as E, with fat scores F hypothesised to be a cause of variation in body mass M. Sex S being genetically determined and therefore subject to Mendelian randomisation[33], and all treatments being randomised with respect to sex, S was considered causally prior to reproductive status R, body mass M, body fat F and parasite burdens P. Reproductive status R being driven by sex, it was considered causally prior to body mass M and parasite burdens P. Vaccine efficacy E was hypothesised to be potentially affected by all other variables. Each node also had its own error term to capture all unobserved sources of variation, all assumed to be independent and identically distributed.

We then tested the validity of the causal hypotheses described above using the DAG to construct each model. For example, the DAG in Fig. 3 implies that parasite infection should be conditionally independent of vaccination if we adjust for diet, reproductive status, sex, and habitat ($P \perp\!\!\!\perp V \mid D, R, S, H$); we therefore fit a binomial generalised linear model $glm(P \sim \alpha + \beta_V V + \beta_D D + \beta_R R + \beta_S S + \beta_H H)$. This process was repeated for all implied independencies derived from the causal graph (see Model validation methods), and the DAG updated until all causal paths were supported by the observed data and consistent with plausible causal expectations. This resulted in the DAG shown in Figure 3.

Average, direct, and indirect causal effects of immunisation on vaccine responsiveness

We fully parametrised $\mathfrak{C}$ (Model 1) using a set of Bayesian linear models containing variables required to ensure de-confounded estimation of the total and natural direct causal effects operating between the nodes of $\mathfrak{C}$ (Fig. 4). Vaccination resulted in an average causal effect of 1.51 $\pm$ 0.11 ($log10 + 1$)-transformed OD units ($logOD$) from unvaccinated animals in this experiment, 1.49 $\pm$ 0.11 of which were estimated as direct (or mediated by unobserved variables, e.g. immune pathways) with a negligible fraction (<0.1 on average) mediated by reproductive status R, body mass M, and body fat F (Fig. 4B). However, vaccine responsiveness E was strongly influenced by both the wild habitat and dietary supplementation. The wild habitat had a direct negative impact of 0.64 $\pm$ 0.17 $logOD$ on vaccine responsiveness, while dietary supplementation resulted in an average reduction of 0.28 $\pm$ 0.12 $logOD$.

Furthermore, while wild wood mice were lighter (-1.3 $\pm$ 0.2 g on average) and accumulated less fat (by 2.3 points on 6-point scale), dietary supplementation increased body mass (+2.53 $\pm$ 0.3 g) though did not affect body fat scores. Interestingly, the effect of dietary supplementation on vaccine responsiveness E was mostly direct (or mediated by unobserved variables) and not mediated by its effects on body mass, fat stores, reproductive status R, or parasite burden P. The total causal effect of sex on E indicated greater vaccine responsiveness in females (0.26 $\pm$ 0.15 $logOD$). However, only a little over half of the effect of sex (0.15 $\pm$ 0.10 $logOD$) was direct, with factors including body mass and fat storage mediating the rest (Fig. 4).

[SAB] Make fig 4B H -> E' and plot on one full page

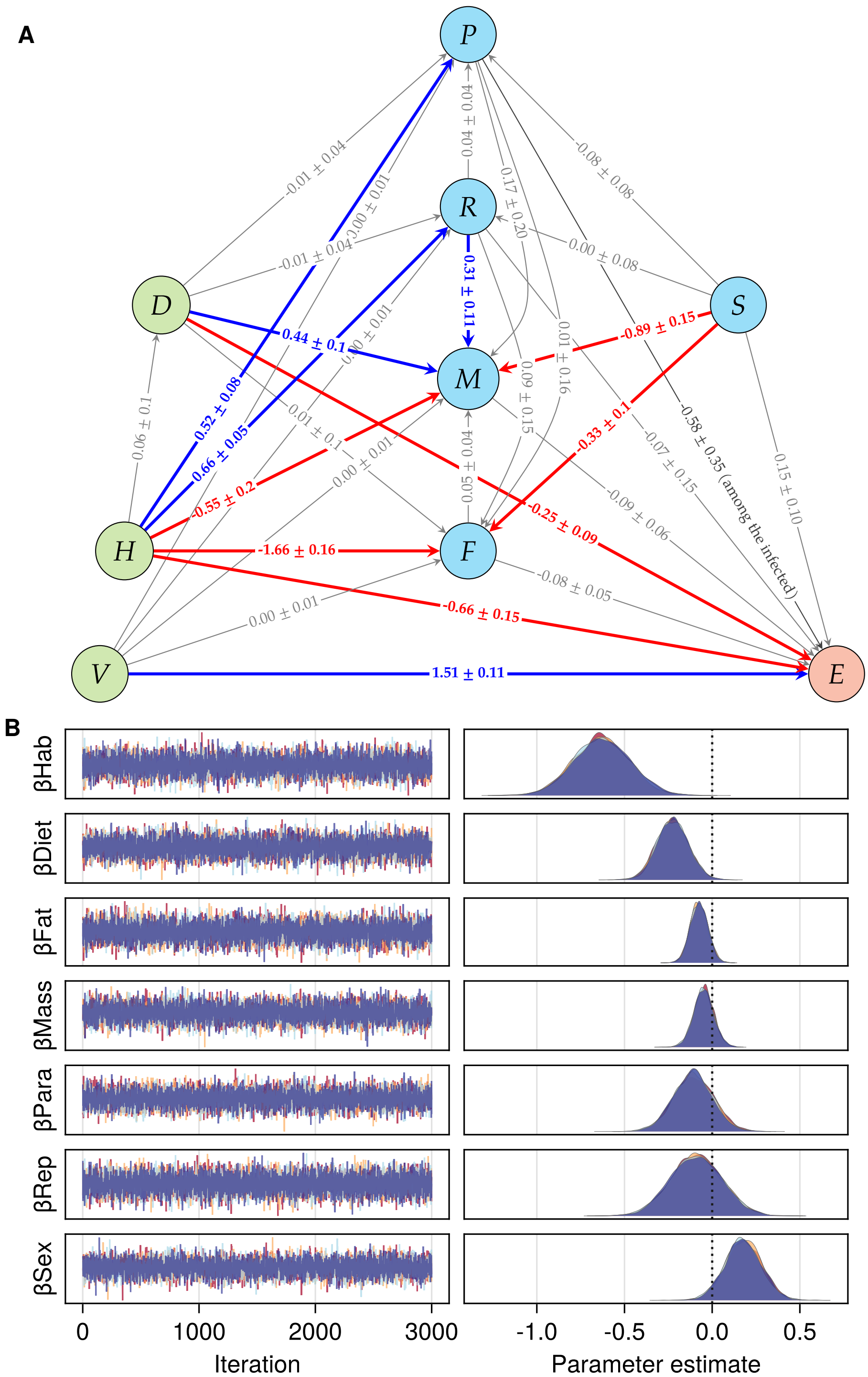


Fig. 4: Fully resolved Causal Model of vaccine efficacy. A, Directed acyclic graph representing point estimates of positive (blue) and negative (red) direct causal effects ± std driving vaccine responsiveness E. **B**, MCMC sampling of posterior distributions of the coefficients estimating the direct effect of the wild habitat H on vaccine responsiveness E, with the adjustment set including, Diet D, Fat scores F, body mass M, parasite burden P, active reproductive status R, and female sex S; vaccine formulation V and individual treated as random effects: $E := \beta_{Hab}H + \beta_{Diet}D + \beta_{Fat}F + \beta_{Mass}M + \beta_{Para}P + \beta_{Rep}R + \beta_{Sex}S$ (see Model parameterisation methods). Note that each edge in panel A shows the direct causal effect of its parent node on its child node, which is conditional on the specific adjustment set for that model; thus, only the coefficient for the edge between D and E is a causal estimate in panel B, while the others are only associative.

Average causal effect of diet supplementation on vaccine responsiveness

Under diet supplementaion, mice in both the wild and the lab ...

Effect of parasite infection among the infected

When considering the whole population including uninfected mice, *H. polygyrus* burden had no clear effect on DT vaccine responsiveness (-0.13 $\pm$ 0.11 $logOD$). However, because parasite infections are overdispersed, we also estimated the causal effects of infection burden on DT vaccine responsiveness only among the infected wild wood mice. The direct effect of parasite burden on vaccine responsiveness among the infected was weakly negative (-0.58 $\pm$ 0.35 $logOD$), when adjusting for diet, sex, reproductive status, and vaccination.

DISCUSSION

The physiological variables measured mediated very little of the effect of immunisation on vaccine responsiveness, if any. However, environmental covariates of supplemented diet and wild habitat had substantial negative effects on vaccine responsiveness.

The 'wild habitat' captures many variables that would still need to be identified - these might include weather, predation, competition, and other stressors that are absent from laboratory conditions.

Effect of nutrition was unexpected. See[13]

Weak / variable effect of worms[9,10,34].

Limitations & Perspectives. We made the simplifying assumptions that:

- parametric assumptions and linearity of the causal effects. Future work will need to generalise the functional form of the causal effects to allow for non-linear effects of the variables on vaccine responsiveness.

- no unmeasured confounders. Future work will need to relax these assumptions to allow for more complex causal structures and to account for unmeasured confounders and mediation.
- random vaccine and diet treatments: we should allow for vaccine and diet uptake (propensity scores) to vary among the treated, as especially in the wild population, these are not strictly enforced.

MATERIALS, METHODS, AND MODELS

All animal work was conducted in accordance to UK Home Office in compliance with the Animals (Scientific Procedures) Act, 1986. Both laboratory and field experiments were approved by the University of Edinburgh Ethical Review Committee and carried out under the UK Home Office Project Licence 70/8543. The dosage of diphtheria vaccine was trialled in laboratory reared wood-mice before the experiments for safety and efficacy. Animal sacrifice was performed using appropriate Schedule 1 Methods. Fieldwork was carried out with permission of the Forestry Commission Scotland under the permit SUR09.

See Supp methods

[ABP] What do we plan to include in the supplemenatry methods?

Laboratory wood mouse experiment

The laboratory experiment was conducted on a wild-derived but now captive, outbred wood mice colony that has been bred and maintained at the University of Edinburgh[32] (See SI for more details). The experiment was carried out in two replicate blocks, with 36 animals in each block (18 males, 18 females). At 12 days before the start of the experiment (d-12), all mice were shifted to new diet regimes and given time to acclimatise; half of the animals in each block (9 males and 9 females) were given TransBreed™, a high-quality whole-diet mouse chow. We have previously shown that wood mice supplemented with TransBreed™ were more resistant to the gastrointestinal nematode *Heligmosomoides polygyrus*, cleared worms more effectively after anthelmintic treatment and produced

[SAB] @ABP, @SV: Check whether mice diet groups are mislabelled against original data.

stronger general (total IgA) and parasite-specific (IgG) immune responses in both wild and laboratory conditions[32]. The other half of the mice in this experiment were fed RM1 (Rat and Mouse Maintenance), a lower quality diet, but commonly used maintenance mouse chow which supports maintenance. Both TransBreed^{TM} and RM1 are standard laboratory diets, with TransBreed^{TM} containing higher amounts of starch, protein, fat and micro-nutrients compared to RM1 (see SI for detailed nutritional compositions). Within each diet regime, both male and female animals were randomly assigned to three experimental groups (Fig. 1). Animals belonging to Groups 1 & 2 (n = 24) were subcutaneously injected with 100µl of diphtheria vaccine (2 Lf per 100µl of adjuvant (9% Potassium Aluminium Sulphate) on d0. The vaccine was prepared in our laboratory one day before administration, using vaccine-grade diphtheria inactivated toxoid protein (Alpha Diagnostic International) adsorbed to the adjuvant (9% Alum). Animals assigned to Group 3 (n = 12) were injected with a control, referred to as 'adjuvant'; 100µl of adjuvant (9% Potassium Aluminium Sulphate) suspended in 1X PBS on d0. Small volume blood samples were taken via tail snip 14 and 17 days after the first injections (d14 and d17) to measure the primary antibody response. Twenty-one days (d21) after the first injections, animals in Group 1 (n = 12) were injected again with 100µl of the diphtheria vaccine (referred to as the 'booster'), whereas those assigned to Group 2 (n= 12) were injected with the control dose as described above (referred to as 'adjuvant'). Animals assigned to Group 3 (n = 12) that had previously been injected with the control dose, now received an injection of 100µl diphtheria vaccine. Another blood sample (20- 50µl) was taken a day later (d22) via cheek venipuncture. All animals were sacrificed 14 days after the second injection (d35) and a blood sample for measuring the secondary antibody response was taken. The same procedures and timeline were carried out for the second experimental block of animals (n = 36). Mice were co-housed throughout the experiment, in same sex groups of three, with equal representation of the three experimental groups in each cage.

Wild wood mice experiment

We conducted a 10-week field experiment in a natural population of wood mice with a design similar to that of the laboratory experiment. The experiment took place in a woodland in Falkirk, Scotland, UK (Callendar Wood, 55.990470, -3.766636), where we established four trapping grids (60m X 40m per grid) with 10m spacing between each of the 35 trapping stations. At each station, we set a pair of Sherman live traps (H.B. Sherman 2X2.5X6.5-inch folding trap, Tallahassee, FL, USA). We randomly selected two of the four grids to be 'supplemented'; 12 days before we began live-trapping, we started supplementing these grids with extra resources. Specifically, we scattered 6 kg TransBreed^{TM} (high quality, whole- diet mouse food) pellets evenly around the grid, and this supplementation continued throughout the experiment, twice per week. The two 'control' grids did not receive any supplementation. Wood mice on all four grids had access to their normal diet and food items, so this extra resources on the 'supplemented 'grids served as an additional high nutritional food resource.

From July-September 2018, wood mice were trapped 2–3 nights per week using live traps baited with grains, carrots, and bedding. Additionally, traps on the supplemented grids were also baited with 1-2 TransBreed^{TM} pellets. For identification, all newly captured mice weighing >13g were subcutaneously tagged with a unique 9-digit microchip passive induced transponder (PIT tag; FriendChip AVID2028, Norco, CA, USA). On first capture, mice were assigned to one of the three vaccine treatment groups described above in the laboratory experiment; mice assigned to these groups received the same primary and booster doses of the diphtheria vaccine or control injections as described below (see Figure 1). Throughout the experiment, weekly small volume blood samples were taken via tail snip from each tagged animal to measure their antibody response to the diphtheria immunisation. Animals recaptured 14 or more days after their second injection were sacrificed and a terminal bleed was collected to measure their antibody response and adult *Heligmosomoides polygyrus* worm burdens. *H. polygyrus* is a natural gastrointestinal nematode found at high prevalence (20-100 %) in wild populations of *A. sylvaticus*[32,35–37], as well as being a well-studied model system of human gastrointestinal nematodes where it has been found to be highly immunomodulatory[38,39].

Additionally, at all captures, we took the following demographic metrics for each individual: sex, body condition, reproductive condition, weight (g), and length (mm). Sex and reproductive condition of each mouse was assigned by examining the genitals, with males having larger urogenital gap compared to females. Males were classified as reproductively active if their testes had visibly descended to scrotal sacs, and females were classified as reproductively active if they had a perforated vagina or were visibly pregnant or lactating. Body condition was measured by assigning dorsal and pelvic fat scores on a qualitative scale from 1-5, where 1 represented a very low condition mouse, while a 5 represented a mouse with ample fat reserves. This was accrued out by palpating the back and pubic bones[40].

Vaccine responsiveness

Vaccine responsiveness was measured as the concentration of anti-diphtheria IgG1 antibodies; this is the gold standard method used in vaccinology to measure the antibody-specific response to immunisation and is used specifically for the diphtheria toxoid vaccine used here[41]. To measure diphtheria vaccine responsiveness, we collected blood samples from both the laboratory and the wild mice and these were centrifuged (@24,000 rpm for 10 min) within 4–6 hours of collection. The sera were separated from the pellets and stored at -80 °C until further analysis. We used ELISAs adapted for diphtheria-specific antibodies from Clerc et al. (2019a), to quantify Anti-diphtheria IgG1 antibodies in the sera samples. 96-well plates (NuncTM MicroWellTM) were coated overnight at 4 °C with diphtheria toxin (2µg/mL) diluted in carbonate buffer (50 µl per well). After washing the plates with TBS (10 X) and Tween80 thrice, 100 µl of TBS(1X)-4percent BSA was added per well and incubated at 37 °C for 2 hours to block non-specific binding sites. Plates were then tapped dry and 50μl of serum samples serially diluted (starting dilution 1:100) in TBS (1X)-4percent BSA buffer were added per well and left at 4 °C for binding overnight. The next day, 50μl HRP-conjugated anti-mouse IgG1 detection antibody (Southern BioTech) was added per well after washing the plates 4 times with TBS (10X) and Tween80 and tapping them dry. After incubation at 37 °C for 1 hour, the plates were washed again, first 4 times with TBS (10X) and Tween80 and then twice with dH20. 50μl of TMB substrate solution was added per well and the enzymatic reaction was left to develop in the dark for 7 min. The reaction was stopped after 7 min using 50μl sulphuric acid (0.18 M) per well. Absorbance at 450 nm was recorded using an ELISA plate reader (Multiscan, Ascent Labsystems) immediately after. For each sample, blank-centred optical densities (OD) of three consecutive wells (of dilutions 1:3200, 1:6400, 1:12800) were averaged to obtain diphtheria-specific IgG antibody concentration.

Data processing and analyses

DT-specific antibody ELISA optical densities (E) were $1 + log_{10}$-transformed and, body weight (M), fat scores (F) and days post-vaccination ($T \geq 1$), were Z score-transformed to help the selection of sensible priors and to improve computational stability. All models were adjusted for time T between vaccination and blood collection to account for natural kinetics of the antibody response. Fat scores ranged from 2 to 5 and were derived from the sum of two five-point scales for pelvic and dorsal fat, where 1 is 'bony' to palpation and 5 is 'fat', and were treated as continuous for the analysis. All other variables were treated as binary.

All data processing was performed in Julia v1.9[42] using packages CSV.jl v0.10[43] and DataFrames.jl v1.3[44].

Generalised linear mixed models of the effects of experimental interventions on habitat, diet, and vaccine formulation

Generalised linear mixed models (GLMM) were used to estimate the effects of habitat, diet and vaccine formulation on vaccine-specific antibody concentrations measured as optical densities read-outs. Mouse ID was modelled as a random effect to account for multiple testing. The interactions between vaccine regimen and habitat, and between diet and habitat were included in the full model to test whether mice in the wild would respond differently to vaccine timing and boosting or to diet supplementation than laboratory mice. Likelihood ratio tests (LRT) were used to test the contribution of the interaction terms to the model. We used Julia v1.9[42] package MixedModels.jl v4.13[45] was used for GLMMs and LRTs.

We used Bayesian multilevel models to minimise the effects of imbalance on our analyses and to quantify the uncertainty of those estimates. Priors for intercepts and slopes were chosen to conform to the observed data, with equal means and 2.5 × the observed variance (see Supplementary Data: Multilevel Models). Typically, intercepts were given Gaussian priors of mean $\mu = mean(y)$ and $2.5 \times \sigma$ those of the observed response variable y while population-level coefficients were given multivariate normal (Gaussian) priors of mean $\mu = 0$ and variance $\sigma = 2$. Population-level residuals were assumed to follow an exponential distribution with scale $\theta = 1/std(y)$. Parameters for group level intercepts and coefficients were given Gaussian priors and their variances were given positive-truncated Cauchy priors with location $x_0 = 0$ and scale $\gamma = 2$. Posterior estimates were sampled using the Hamiltonian Monte Carlo algorithm No U-Turn Sampler[46], typically across 4 threads simultaneously with $> 3,000$ draws. Turing.jl[47] was used for Bayesian modelling; Makie.jl v0.17[48] was used for plotting.

Structural Causal Models

The structural causal model (SCM) framework[29,49] proposes explicit qualitative solutions to biases inherent in datasets and can guide the statistical estimation of the mechanisms by which data are generated in complex systems[50,51]. Further, Bayesian statistical methods for estimating the causal effects indicated by the SCM allows flexibility of imbalanced small data and missing values[52], and provides explicit estimation of uncertainty given assumptions made by the model[53]. Specifically, here we have made three important assumptions: vaccine, diet, and habitat treatments are random; there are no (statistically relevant) unmeasured confounders; and all relationships between variables are linear.

Model construction

Each of the variables X described above was included in the structural causal model $\mathfrak{C}$ with the following assignments of the form $X_j := f(X_{k \neq j}) + \mathcal{U}_j$ where the unmeasured noise variable $\mathcal{U}_j \sim Normal(0, 4)$, was assumed to be independent and identically distributed, and $f()$ was assumed to be linear (Supplementary Data: SCM construction):

A directed acyclic graph (DAG) expressing $\mathfrak{C}$ (Model 1) was used to build unconfounded models for the estimation of causal effects between each variable (Figure 3).

Model construction was performed in Julia v1.9[42] using package Dagitty.jl and Turing.jl[47], and the DAG corresponding to $\mathfrak{C}$ was drawn in Tikz[54].

Model validation

To validate $\mathfrak{C}$ (Model 1), we tested the independencies it implies following chain, fork, and collider rules[29,55]. Specifically, we used the observed values for each of the nodes of the causal graph (Figure 3) to test the truth of the following 13 implied independences: $(D \perp\!\!\!\perp S)$, $(D \perp\!\!\!\perp T \mid H)$, $(D \perp\!\!\!\perp V)$, $(F \perp\!\!\!\perp V \mid T, H)$, $(H \perp\!\!\!\perp S)$, $(H \perp\!\!\!\perp V)$, $(M \perp\!\!\!\perp V \mid T, H)$, $(P \perp\!\!\!\perp T \mid D, R, S, H)$, $(P \perp\!\!\!\perp V \mid T, H)$, $(P \perp\!\!\!\perp V \mid D, R, S, H)$, $(R \perp\!\!\!\perp V \mid T, H)$, $(S \perp\!\!\!\perp T)$, and $(S \perp\!\!\!\perp V)$. For each of these tests, we constructed a generalised linear mixed model with the appropriate link function, with mouse ID and immunisation protocol as random effects, and considered the statement true if $P_X > 0.1$, where $X \subseteq \mathfrak{C}$ (Supplementary Data: SCM construction). We used MixedModels.jl v4.13[45] to test the conditional independences implied by $\mathfrak{C}$.

Model parameterisation

Next, we encoded $\mathfrak{C}$ as a Bayesian network in Turing.jl[47] to represent each variable as a probability distribution to be updated by the observed values. This allowed us to capture conditional expected values for each variable and their uncertainty. Posterior estimates were validated with MixedModels.jl[45].

Missing data. Twenty-two mice had no fat scores, resulting in 67 missing values in the full repeated sampling dataset. To avoid having to remove those mice from the analysis, we used Bayesian imputation to estimate plausible probability distributions of each of the missing values in light of its parents (PA) in $\mathfrak{C}$ (Model 1), where $F := f(PA_F) = f(T, H, D, P, S)$ (Figure 3).

$\dot{F}$ denotes the variable F that contains imputed values; E_i denotes the expected distribution of E given $\dot{F}$.

Model updating and sampling. All Bayesian modelling was performed in Julia v1.9[42] using package Turing.jl v0.25[47].

ACKNOWLEDGEMENTS

Add details of the ISSF vaccination grant, Saudamini's Darwin Trust funding, ABPs Chancellors Fellowship, Amy Sweeny's Darwin Trust funding, SAB RS and BMGF. - what else?

COMPETING INTERESTS

The authors declare that there are no competing interests.

AUTHOR CONTRIBUTIONS

Conceived and designed the experiments: ABP, SAB

Performed the experiments: SV, AS, ABP, SAB

Analysed the data: SV, SAB

Contributed reagents/materials/analysis tools: ABP, SAB

All the authors reviewed the manuscript.

REFERENCES

1. Greaney, A. *et al.* Mapping mutations to the SARS-CoV-2 RBD that escape binding by different classes of antibodies. *Nat Commun* **12,** 4196 (2021).
2. Harvey, W. *et al.* SARS-CoV-2 variants, spike mutations and immune escape. *Nat Rev Microbiol* **19,** 409–424 (2021).
3. Wang, Z. *et al.* mRNA vaccine-elicited antibodies to SARS-CoV-2 and circulating variants. *Nature* **592,** 616–622 (2021).
4. Kennedy, D. & Read, A. Why the evolution of vaccine resistance is less of a concern than the evolution of drug resistance. *Proc. Natl. Acad. Sci. U S A* **115,** 12878–12886 (2018).
5. Posteraro, B. *et al.* The link between genetic variation and variability in vaccine responses: systematic review and meta-analyses. *Vaccine* **32,** 1661–1669 (2014).
6. Flanagan, K., Fink, A., Plebanski, M. & Klein, S. Sex and Gender Differences in the Outcomes of Vaccination over the Life Course. *Annu Rev Cell Dev Biol* **33,** 577–599 (2017).
7. Shah, N., Imami, N., Kelleher, P., Barclay, W. & Johnson, M. Pregnancy-related immune suppression leads to altered influenza vaccine recall responses. *Clin Immunol* **208,** 108254 (2019).
8. Elias, D., Britton, S., Aseffa, A., Engers, H. & Akuffo, H. Poor responsiveness of BCG in helminth infected population is associated with increased in vitro TGF-beta production. *Vaccine* **26,** 3897–3902 (2008).
9. Wait, L., Dobson, A. & Graham, A. Do parasite infections interfere with immunisation? A review and meta-analysis. *Vaccine* **38,** 5582–5590 (2020).
10. Natukunda, A. *et al.* The effect of helminth infection on vaccine responses in humans and animal models: A systematic review and meta-analysis. *Parasite Immunol.* **44,** e12939 (2022).
11. Huda, M. *et al.* Bifidobacterium Abundance in Early Infancy and Vaccine Response at 2 Years of Age. *Pediatrics* **143,** e20181489 (2019).
12. Saleem, A. *et al.* responsiveness of poliovirus vaccines in chronically malnourished infants: a randomized controlled trial in Pakistan. *Vaccine* **33,** 2757–2763 (2015).
13. Yap, P., Utzinger, J., Hattendorf, J. & Steinmann, P. Influence of nutrition on infection and re-infection with soil-transmitted helminths: a systematic review. *Parasit Vectors* **7,** 229 (2014).
14. Fourati, S. *et al.* Pan-vaccine analysis reveals innate immune endotypes predictive of antibody responses to vaccination. *Nat. Immunol.* (2022).
15. Vazquez-Prokopec, G. *et al.* Coupled Heterogeneities and Their Impact on Parasite Transmission and Control. *Trends Parasitol* **32,** 356–367 (2016).
16. Van Dorst, M. *et al.* Immunological factors linked to geographical variation in vaccine responses. *Nat Rev Immunol* (2023).
17. Pronker, E., Weenen, T., Commandeur, H., Claassen, E. & Osterhaus, A. Risk in vaccine research and development quantified. *PLoS One* **8,** e57755 (2013).
18. Schaid, D. *et al.* Heritability of vaccine-induced measles neutralizing antibody titers. *Vaccine* **35,** 1390–1394 (2017).

19. Boahen, C. *et al.* A functional genomics approach in Tanzanian population identifies distinct genetic regulators of cytokine production compared to European population. *Am J Hum Genet* **109,** 471–485 (2022).

20. Voigt, E. *et al.* Genetically defined race, but not sex, is associated with higher humoral and cellular immune responses to measles vaccination. *Vaccine* **34,** 4913–4919 (2016).

21. Zimmermann, P. & Curtis, N. Factors That Influence the Immune Response to Vaccination. *Clin Microbiol Rev* **32** (2019).

22. Kabagenyi, J. *et al.* Urban-rural differences in immune responses to mycobacterial and tetanus vaccine antigens in a tropical setting: A role for helminths. *Parasitol Int* **78,** 102132 (2020).

23. Babayan, S. *et al.* Wild immunology: converging on the real world. *Ann N Y Acad Sci* **1236,** 17–29 (2011).

24. Beura, L. *et al.* Normalizing the environment recapitulates adult human immune traits in laboratory mice. *Nature* **532,** 512–516 (2016).

25. Graham, A. Naturalizing mouse models for immunology. *Nat Immunol* **22,** 111–117 (2021).

26. Rosshart, S. *et al.* Wild Mouse Gut Microbiota Promotes Host Fitness and Improves Disease Resistance. *Cell* **171,** 1015–1028.e13 (2017).

27. Rosshart, S. *et al.* Laboratory mice born to wild mice have natural microbiota and model human immune responses. *Science* **365,** eaaw4361 (2019).

28. Mangino, M., Roederer, M., Beddall, M., Nestle, F. & Spector, T. Innate and adaptive immune traits are differentially affected by genetic and environmental factors. *Nat Commun* **8,** 13850 (2017).

29. Pearl, J. *Causality* (Cambridge University Press, 2009).

30. Glymour, C., Zhang, K. & Spirtes, P. Review of Causal Discovery Methods Based on Graphical Models. *Front Genet* **10,** 524 (2019).

31. Bareinboim, E., Correa, J. D., Ibeling, D. & Icard, T. On Pearl's Hierarchy and the Foundations of Causal Inference Probabilistic and Causal Inference: The Works of Judea Pearl, 507–556 (2022).

32. Sweeny, A. *et al.* Supplemented nutrition decreases helminth burden and increases drug efficacy in a natural host-helminth system. *Proc Biol Sci* **288,** 20202722 (2021).

33. Lawlor, D. A., Harbord, R. M., Sterne, J. A. C., Timpson, N. & Davey Smith, G. Mendelian randomization: Using genes as instruments for making causal inferences in epidemiology. *Statistics in Medicine* **27,** 1133–1163 (2008).

34. Brückner, S. *et al.* Effect of Antihelminthic Treatment on Vaccine responsiveness to a Seasonal Influenza Vaccine in Primary School Children in Gabon: A Randomized Placebo-Controlled Trial. *PLoS Negl Trop Dis* **9,** e0003768 (2015).

35. Gregory, R., Keymer, A. & Clarke, J. Genetics, sex and exposure: the ecology of *Heligmosomoides polygyrus* (Nematoda) in the wood mouse. *The Journal of Animal Ecology* (1990).

36. Keymer, A. & Dobson, A. P. The ecology of helminths in populations of small mammals. *Mammal Review* **17,** 105–116 (1987).

37. Knowles, S. *et al.* Stability of within-host-parasite communities in a wild mammal system. *Proc Biol Sci* **280,** 20130598 (2013).

38. Maizels, R. *et al.* Immune modulation and modulators in *Heligmosomoides polygyrus* infection. *Exp Parasitol* **132,** 76–89 (2012).

39. Reynolds, L., Filbey, K. & Maizels, R. Immunity to the model intestinal helminth parasite *Heligmosomoides polygyrus*. *Semin Immunopathol* **34,** 829–846 (2012).

40. Ullman-Culleré, M. & Foltz, C. Body condition scoring: a rapid and accurate method for assessing health status in mice. *Lab Anim Sci* **49,** 319–323 (1999).

41. Faruq, A., Dadson, L., Cox, H., Alcock, F. & Parker, A. Technical and diagnostic performance of five commercial anti-diphtheria toxoid IgG enzyme-linked immunosorbent assay kits. *Clin Vaccine Immunol* **17,** 1612–1616 (2010).

42. Bezanson, J., Edelman, A., Karpinski, S. & Shah, V. B. Julia: A fresh approach to numerical computing. *SIAM review* **59,** 65–98 (2017).

43. Quinn, J. *CSV.jl* version v0.10.4. May 2022. `https://github.com/JuliaData/CSV.jl`.

44. White, J. M. *DataFrames.jl* version v1.3.4. May 2022. `https://github.com/JuliaData/DataFrames.jl`.

45. Bates, D. *et al. JuliaStats/MixedModels.jl: v4.6.3* version v4.6.3. May 2022. `https://doi.org/10.5281/zenodo.6513029`.

46. Hoffman, M. D. & Gelman, A. The No-U-Turn sampler: adaptively setting path lengths in Hamiltonian Monte Carlo. *J. Mach. Learn. Res.* **15,** 1593–1623 (2014).

47. Ge, H., Xu, K. & Ghahramani, Z. *Turing: a language for flexible probabilistic inference* in *International Conference on Artificial Intelligence and Statistics, AISTATS 2018, 9-11 April 2018, Playa Blanca, Lanzarote, Canary Islands, Spain* (2018), 1682–1690. `http://proceedings.mlr.press/v84/ge18b.html`.

48. Danisch, S. & Krumbiegel, J. Makie.jl: Flexible high-performance data visualization for Julia. *Journal of Open Source Software* **6,** 3349 (2021).

49. Peters, J., Janzing, D. & Schölkopf, B. *Elements of causal inference: foundations and learning algorithms* (MIT press, 2017).

50. Bareinboim, E. & Pearl, J. Causal inference and the data-fusion problem. *Proc Natl Acad Sci U S A* **113,** 7345–7352 (2016).

51. Pearl, J. On measurement bias in causal inference. *arXiv* (2012).

52. McElreath, R. *Statistical Rethinking: A Bayesian Course with Examples in R and STAN* 594 (CRC Press, 2020).

53. Gelman, A. *et al. Bayesian Data Analysis, Third Edition* 675 (CRC Press, 2013).

54. Tantau, T. *Graph Drawing in TikZ* in *Proceedings of the 20th International Conference on Graph Drawing* (Springer-Verlag, Redmond, WA, 2013), 517–528. ISBN: 978-3-642-36762-5.

55. Cinelli, C., Forney, A. & Pearl, J. A Crash Course in Good and Bad Controls (2021).

TABLES

Table 1: Posterior predictions of model parameters for random intercept Gaussian model $E \sim Normal(\alpha[ID] + H + D + V + V : H, \sigma)$.

Parameter	Mean	Std	5%	50%	95%
Intercept	1.02	0.85	-0.39	1.032	2.43
Habitat	-0.519	0.806	-1.863	-0.513	0.795
Diet	-0.253	0.072	-0.373	-0.253	-0.137
A	0.0	0.871	-3.307	-1.892	-0.438
D	2.062	0.856	-1.228	0.175	1.598
AD	1.256	0.875	-2.061	-0.651	0.84
DA	3.652	0.87	0.343	1.751	3.215
DD	2.996	0.865	-0.304	1.091	2.534
A:H	0.0	0.818	-0.864	0.47	1.831
D:H	-0.594	0.811	-1.434	-0.125	1.223
AD:H	-0.371	0.823	-1.244	0.103	1.472
DA:H	-1.348	0.818	-2.228	-0.874	0.496
DD:H	-0.527	0.818	-1.389	-0.061	1.31

Environmental drivers of low vaccine responsiveness in a lab-to-wild mouse model

Simon A. Babayan[1,†,*], Saudamini Venkatesan[2,†], Jessica Hall[1,2], Ewan Smith[1], Amy Sweeny[2], and Amy B. Pedersen[2,*]

[1]School of Biodiversity, One Health & Veterinary Medicine, University of Glasgow, Glasgow, UK
[2]Institute of Ecology and Evolution, School of Biological Sciences, University of Edinburgh, Edinburgh, UK
[†]These authors contributed equally to this work
[*]Corresponding authors: simon.babayan@glasgow.ac.uk, amy.pedersen@ed.ac.uk

10th January 2024

SUPPLEMENTARY INFORMATION

Supp methods

Insert additional details about the lab mouse colony details of the diet make up of trans-breed and RM1

Supplementary Data: Multilevel Models

Supplementary Data: SCM construction

Supplementary Data: SCM specification

Supplementary Data: Wood mouse data

Supplementary Figures

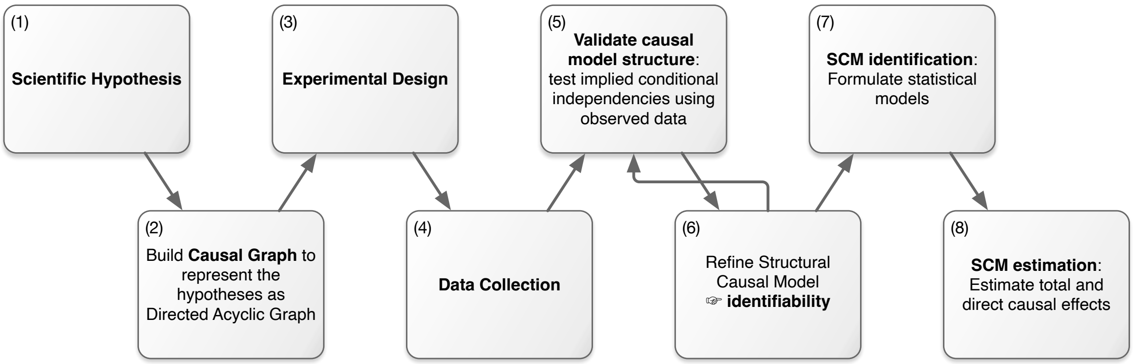


Fig. S1: Flow diagram of the SCM. The Structural Causal Model (SCM) process explicitly represents scientific hypotheses (1) as a directed acyclic graph (DAG). This DAG is mathematically encoded as a set of nested equations that describe the flow of causation between variables, and helps inform lab and field experimental design (3) and data collection (4). After processing, the data are used to test the validity of the causal assumptions underlying the SCM, e.g. by testing the conditional independencies implied by the DAG (5). If the assumptions are not supported, refinements of the SCM (6) are necessary. When all conditions are met, the SCM is identifiable and statistical models can be formulated to adjust for confounding (7). Parameters from these models can then be used as estimates of direct and indirect causal effects (8).

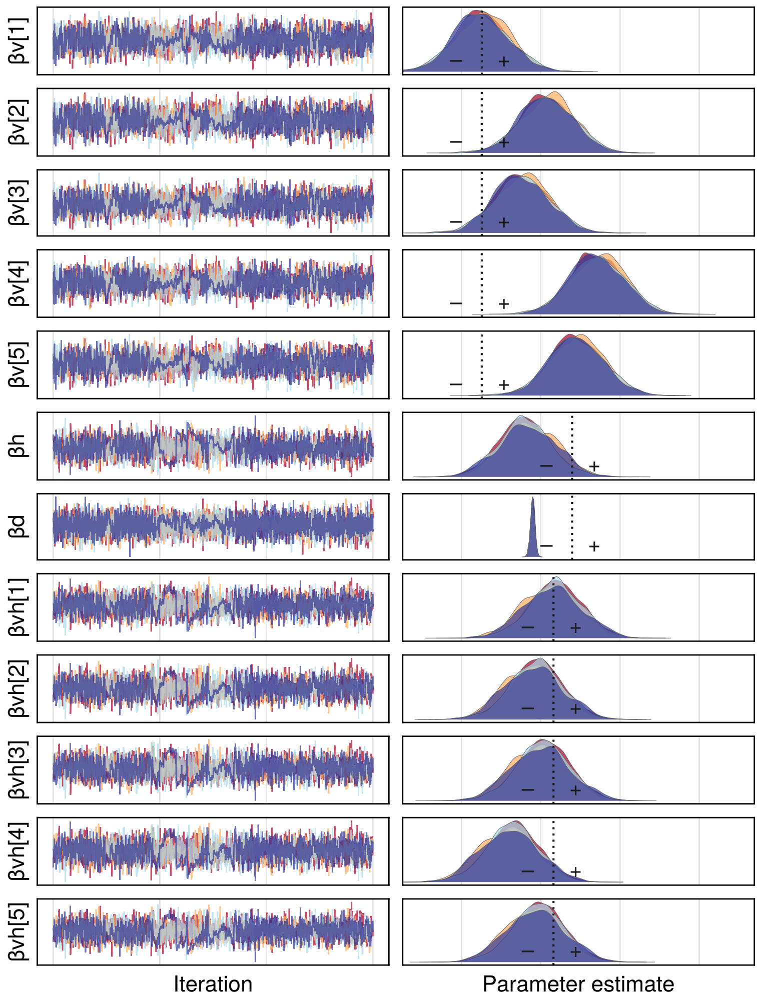


Fig. S2: MCMC chains and posterior distributions of coefficients for Diet, Habitat, Time post immunisation, and vaccine formulations A, AD, D, DA, and DD.